

Resilience in the Workplace: A Multilevel Review and Synthesis

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In an increasingly dynamic business environment, the concept of resilience is fundamental to understanding how employees successfully handle adversity. Yet, the operationalisation of the concept, the factors which lead to its development, and how and why it influences outcomes of interest to organisations are issues still under debate in the literature. In this article, we address these debates by undertaking a critical review of research on resilience in the workplace at both the individual and team levels. We pinpoint different conceptualisations of resilience and highlight how resilience has been measured in extant quantitative work. Further, we provide a systematic literature review of empirical work on the antecedents and outcomes of resilience at the individual and team levels, as well as conduct a review of work that has introduced resilience as a moderator or mediator. In doing so, we highlight theoretical approaches that have been adopted to study resilience in the workplace. Based on our review of the extant empirical work on resilience, we develop a roadmap for future research. In particular, we pinpoint relevant theoretical approaches that help us understand the mechanisms underlying the development and outcomes of resilience and highlight opportunities for empirical advancement of the literature.

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INTRODUCTION

Resilience is one of the core constructs of positive organisational behaviour (Luthans, 2002), and has been defined as positive adaptation in the face of adversity (Sutcliffe & Vogus, 2003). Given today's disruptive and adverse work environment, scholars' and practitioners' interest in workplace resilience has greatly increased in recent years (King, Newman, & Luthans, 2016; van der Vegt, Essens, Wahlström, & George, 2015). To synthesise this burgeoning body of work, a number of reviews and meta-analyses on resilience in organisational contexts have been undertaken. For example, Linnenluecke (2017) used bibliographic mapping to identify five different streams of research that investigate resilience in and of organisations. Williams, Gruber, Sutcliffe, Shepherd, and Zhao (2017) reviewed work on organisational resilience and elaborated on its linkage to research on crisis management. Kossek and Perrigino (2016) provided an occupation-specific discussion of employees' resilience. Britt, Shen, Sinclair, Grossman, and Klieger (2016) reviewed different conceptualisations of employees' resilience as well as resilience-building programmes. Kuntz, Malinen, and Näswall (2017) focused on how employee resilience can be developed as a behavioural capability. Adding to this, Vanhove, Herian, Perez, Harms, and Lester (2016) as well as Robertson, Cooper, Sarkar, and Curran (2015) reviewed the effectiveness of resilience-building programmes in organisations.

Despite the contribution of these prior reviews and meta-analyses on resilience, important questions are still unanswered. First, we encountered neither a comprehensive review of the antecedents and outcomes of resilience in the workplace at different levels of analysis, nor sufficient scrutiny of the theories used to explain how resilience develops and influences work outcomes. The development of conceptual frameworks for the study of resilience in organisational contexts, such as those provided by Fisher, Ragsdale, and Fisher (2018), Kossek and Perrigono (2016), and Britt, Shen, Sinclair, Grossman, and Klieger (2016), have been an important step to address problems arising from conceptual inconsistencies. However, these integrative frameworks still do not fully map the nomological network of variables to which resilience in the workplace is related. Further, such prior work does not distinguish between research in which resilience is treated as an antecedent, outcome, mediator or moderator at different levels of analysis. Second, prior literature reviews have not addressed the topic of resilience at the team level of analysis. This is unfortunate given that scholars have stressed the need for understanding resilience at the team level (Alliger, Cerasoli, Tannenbaum, & Vessey, 2015; King et al., 2016; Stoverink, Kirkman, Mistry, & Rosen, 2018) and have started to empirically investigate the antecedents and outcomes of team resilience (e.g., Carmeli, Friedman, & Tishler, 2013; Meneghel, Martínez, &

Salanova, 2016b). To synthesise this empirical work and address scholars' calls to advance knowledge of team resilience, the present study also reviews empirical work at the team level of analysis. Finally, although prior research has provided an overview of instruments that have been used to measure resilience among general adult and/or clinical populations (DeSimone, Harms, Vanhove, & Herian, 2017; Pangallo, Zibarras, Lewis, & Flaxman, 2015; Windle, Bennett, & Noyes, 2011), existing work has not examined the instruments that have been used to measure resilience in the workplace context. Yet, as empirical research on resilience in the workplace is flourishing, a better understanding about the differences between various measurement instruments is of value to scholars that aim to investigate this concept in a workplace context.

To address these research gaps, we undertake a comprehensive review of the empirical research conducted on resilience in the workplace. In doing so, we make a number of contributions to the literature. First, going beyond extant reviews and conceptual studies (e.g., Britt et al., 2016; Fisher et al., 2018; Kossek & Perrigono, 2016; Stoverink et al., 2018), we provide an important contribution by synthesising an emerging, yet significant, body of literature on the antecedents and outcomes of resilience in the workplace at different levels of analysis, and by pinpointing literature that has introduced resilience as a moderator or mediator. In doing so, we highlight the theoretical mechanisms that have been used to explain how resilience develops and transmits its effects at the individual and team levels, and we develop an overview of research that highlights the nomological network of resilience at both the individual and team levels of analysis. We make a second key contribution to the literature by reviewing how resilience in the workplace has been measured in previous quantitative research. This is an important contribution, given that no guidelines for measuring resilience in the workplace are currently available. Yet, recent research indicates that resilience may be context-dependent (Kossek & Perrigino, 2016; Todt, Weiss, & Hoegl, 2018; Tonkin, Malinen, Näswall, & Kuntz, 2018). Thus, a better understanding of the rationale underlying different measurement instruments helps scholars to select appropriate measures to study resilience in the workplace. Finally, we develop a comprehensive agenda for future research on resilience in the workplace. Specifically, we highlight theoretical perspectives that might advance our knowledge on how resilience develops and influences important outcomes. In addition, we provide suggestions for further empirical research on resilience in workplace settings. Our recommendations will help scholars design future studies that advance our knowledge as to how resilience in the workplace develops and influences outcomes at different levels of analysis.

LITERATURE REVIEW

The foundation of this paper is a systematic literature review of empirical research on resilience in the workplace. We followed Rousseau, Manning, and Denyer (2008) for best-practice scientific recommendations for conducting literature reviews. In doing so, we undertook a keyword search in Web of Science, Scopus, Google Scholar, and related databases to identify peer-reviewed articles with resilience or derived terms (i.e., resiliency or resilient) in their titles, abstracts, or keywords that were published in print or available online in English by December 2017. Further articles were discovered through a backward and forward citation search of the identified articles and a manual search in key management and organisational psychology journals. In sum, we identified and screened 1,242 manuscripts based on their titles and abstracts. As prior reviews have already focused on organisational resilience (Williams et al., 2017), the second-order construct of psychological capital (PsyCap) (Dawkins, Martin, Scott, & Sanderson, 2013; Newman, Ucbasaran, Zhu, & Hirst, 2014), and the effectiveness of resilience training interventions (Robertson et al., 2015; Vanhove et al., 2016), we adopted the following exclusion criteria: We excluded literature if it (1) did not focus on the workplace or organisational context; (2) did not focus on individual or team resilience; (3) was not empirical; or (4) investigated resilience as an outcome of training interventions. With regard to studies that measured resilience as one dimension of psychological capital (Newman et al., 2014), we included only those studies that reported empirical results for the sub-dimension of resilience (e.g., West, Patera, & Carsten, 2009; Youssef & Luthans, 2007). To ensure the quality of the empirical research in the literature review, we further excluded literature that was not published in a peer-reviewed, ISI-listed journal. Based on these criteria, we identified 183 articles for full-text screening and excluded 100 additional articles. Figure 1 provides a flow chart diagram that illustrates our screening process. Our final sample included 83 peer-reviewed articles, the first of which was published in 1990. Of the identified articles, 65 adopted a quantitative study design while 18 adopted a qualitative study design.

As prior work has not adopted multi-level or cross-level designs, we review work on resilience at the individual and team levels separately. We organise our paper as follows: We start by focusing on resilience at the individual level of analysis and highlight the numerous ways in which individual resilience has been conceptualised before examining how it has been operationalised and measured in prior research on resilience in the workplace. Next, we critically review empirical work on the antecedents and outcomes of resilience and on its use as a moderator or mediator at the individual level of analysis. In

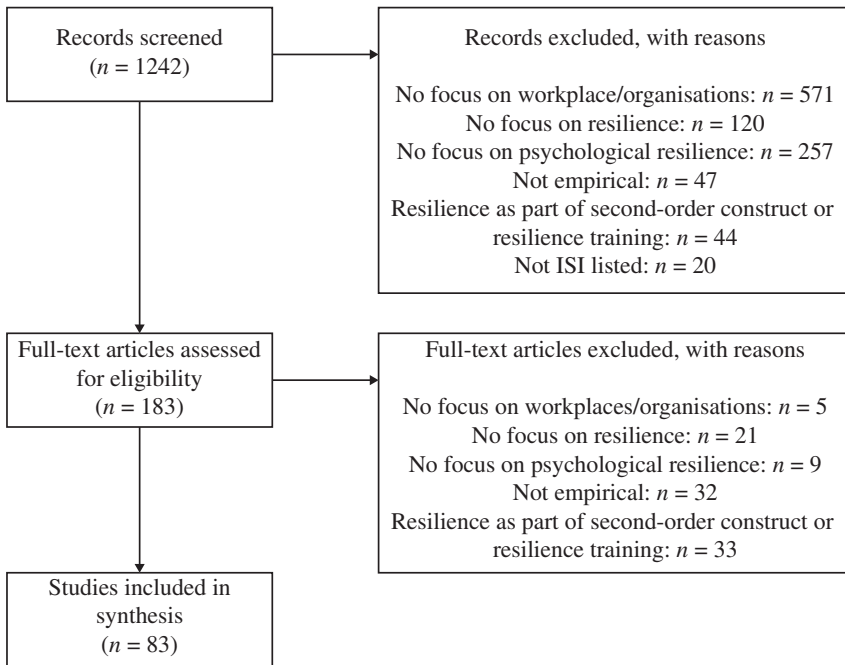


FIGURE 1. Flow diagram of literature screening.

doing so, we highlight the different theoretical perspectives adopted in prior research. In the next part of the paper, we focus on resilience at the team level of analysis. In doing so, we review and discuss conceptualisations, measurement approaches, and empirical research on team resilience in the workplace. Finally, we discuss and critically analyse the findings of our review, highlighting gaps in our extant knowledge from which we derive a comprehensive roadmap to guide future research.

DEFINING RESILIENCE

Research on resilience in the workplace builds on the ideas of positive psychology, which focuses on positive human strengths (Seligman & Csikszentmihalyi, 2000; Luthans, 2002). In the workplace context, resilience has been studied in different industries and professions, such as general business organisations (e.g., Shin, Taylor, & Seo, 2012), healthcare (e.g., Gabriel, Diefendorff, & Erickson, 2011), or the military (e.g., Lee, Sudom, & Zamorski, 2013).

In this manuscript, we define resilience as a “dynamic process encompassing positive adaptation within the context of significant adversity” (Luthar, Cicchetti, & Becker, 2000, p. 543). Resilience encompasses two defining elements (Masten, 2001). The first defining element is the experience of adversity. Fisher et al. (2018) highlight that adversity at work may come in the form of isolated and high-intensity circumstances (e.g., crisis), or may come in the form of lower-intensity but high-frequency or high-duration circumstances (e.g., forms of work stress). In either form, the experience of adversity creates a challenge as well as a threat for an entity’s performance and well-being (Richardson, 2002). The second defining element of resilience is positive adaptation (Masten, 2001), through which the entity returns to a steady state of well-being or performance or even bounces beyond it (Britt et al., 2016; Luthar, Cicchetti, & Becker, 2000).

INDIVIDUAL RESILIENCE IN THE WORKPLACE

Conceptualising Resilience at the Individual Level

The overwhelming majority of research on resilience in the workplace focuses on the individual level of analysis (King et al., 2016). In this section, we briefly highlight the different ways in which individual resilience has been conceptualised in prior work. For further discussion on the conceptualisation of resilience at the individual-level, see Britt et al. (2016), Kossek and Perrigino (2016), and Vanhove, Herian, Harms, and Luthans (2015).

Scholars have conceptualised individual resilience as a stable personality trait, a state-like developable capacity, or a process (Kossek & Perrigino, 2016; Richardson, 2002). The trait perspective conceptualises resilience as a discrete and stable personal characteristic or a bundle of different personal strengths (Wanberg & Banas, 2000). This conceptualisation implies that resilient individuals are generally better able to withstand adversity and setbacks than non-resilient individuals (Shin et al., 2012). The capacity perspective understands resilience as a state-like attribute, which, although stable over certain periods, is also malleable in the long run. As such, under this perspective, resilience is seen as developable via training (Luthans, 2002). The process perspective refers to resilience as an unfolding progression that emerges in response to a number of contingencies and results in the demonstration of positive adaptation (McLarnon & Rothstein, 2013; Moenkemeyer, Hoegl, & Weiss, 2012).

In the present review, we conceptualise resilience as a process. This perspective allows us to take into account temporal and developmental aspects that are important in the study of resilience (Fisher et al., 2018). Further, a process conceptualisation of resilience can account for contingencies (Sutcliffe &

Vogus, 2003) and therefore allows for domain specificity, that is, specification for the workplace context. This is important because recent research suggests that resilience, similar to other psychological constructs, may be domain specific (Harms, Vanhove, & Luthans, 2017; Kossek & Perrigino, 2016; Todt et al., 2018). Finally, this process view of resilience is in line with recent theorising (Fisher et al., 2018; Kossek & Perrigino, 2016), which argues that a combination of actions, malleable capacities, and stable, trait-like elements affects the outcome of the resilience process. Therefore, the process perspective offers a holistic and integrative approach to studying resilience. From a process perspective, resilience outcomes may be influenced by both resilience mechanisms and resilience-promoting factors. According to Fisher et al. (2018), resilience mechanisms can best be understood as those experiences, reactions, and behaviours that individuals apply in the face of adversity, such as certain coping strategies or emotional responses. Resilience-promoting factors refer to personal or environmental characteristics, which are present irrespective of an individual's experience of adversity, but which can buffer the negative effects of adversity or foster resilience mechanisms during adverse experiences. An example of a promoting factor is the character-trait of hardiness (Fisher et al., 2018). We advocate that the experience of adversity, resilience mechanisms, and resilience-promoting factors, as well as resilience outcomes, mark important elements of the resilience process.

Measuring Individual Resilience

In the following section, we discuss the resilience measures that have been used or developed in the articles we reviewed on individual resilience in the workplace. As such, Table 1 provides an overview and discussion of resilience measures that have either been developed to measure individual resilience in a workplace context or have been used in empirical research on individual resilience in the workplace. The table does not include resilience measures that have not been specifically developed for the workplace context and/or have not been used in empirical studies on individual resilience in the workplace (e.g., such as DeSimone, Harms, Vanhove, & Herian, 2017).

The variety of scales reflects the different conceptualisations of individual resilience in the workplace. Scales developed in the 1990s tended to rely on conceptualisations of resilience as stable trait (e.g., Block & Kremen, 1996; Wagnild & Young, 1993). A similar picture emerges for measures of resilience outside workplace contexts, for which Pangallo et al. (2015) reported that scholars predominantly operationalised resilience as a trait-like characteristic. In contrast, the overall majority of scales developed for, or applied in, the workplace context draw on conceptualisations of individual resilience as a capacity (e.g., Connor & Davidson, 2003). Items from these scales

TABLE 1
Overview on Measurement Instruments Developed/Used to Measure Individual Resilience in the Workplace

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Noe et al. (1990)	Career Resilience (subscale of career motivation)	13 items	Construct (ConV; DisV)	Rationale: career resilience as work-related ability Focus: adaption to and coping with changing or negative work situations Context: specific for work careers	
Wagnild and Young (1993)	Resilience Scale (RS)	25 items	Construct (EFA) Criterion (PredV)	Rationale: resilience as stable, positive personal characteristic Focus: (1) personal competence; (2) acceptance of self and life Context: no work focus	Parker et al. (2015) Rice and Liu (2016) (14 items) Sommer et al. (2016) (17 items)
London (1993)	Career Resilience (subscale of career motivation)	7 items	Construct (EFA)	Rationale: career resilience as a trait-like characteristic Focus: maintenance or persistence in career with focus on feelings and attitudes Context: specific for work careers	
Carson and Bedeian (1994)	Career Resilience (subscale of career commitment scale)	4 items	Construct (EFA; ConV; DisV) Criterion (PredV)	Rationale: career resilience as work-related ability Focus: maintenance or persistence in career with focus on attitudes and behaviours Context: specific for work careers	Carless and Bernath (2007) Green et al. (2011) Lyons et al. (2015)
Block and Kremen (1996)	Ego-Resiliency Scale (ER89)	14 items	Construct (ConV; DisV)	Rationale: resilience as a generalised, characterological individual quality Focus: ability to change from and also return to the individual's characteristics level of ego-control Context: no work focus	Shin et al. (2012) van Erp et al. (2015) (4 items) Youssef and Luthans (2007)

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Grzeda and Prince (1997)	Career Resilience (subscale of career motivation)	14 items	Construct (EFA; ConV; DisV)	Rationale: career resilience as work-related ability Focus: maintenance or persistence in career with focus on feelings, attitudes and behaviours Context: specific for work careers	
Gowan et al. (2000)	Career Resilience [based on Waterman Jr, Waterman, & Collard, (1994)]	8 items	Not validated	Rationale: career resilience as a personal quality Focus: (1) flexibility; (2) creativeness; (3) self-reliance; (4) ambition; (5) desire to learn new things; (6) future career plans; (7) confidence; (8) career ownership Context: specific for work careers	
Lounsbury and Gibson (2000)	Emotional Resilience	15 items	Construct (ConV; DisV) Criterion (PredV; IncrV)	Rationale: emotional resilience as personality trait (conceptualised as the inverse of neuroticism) Focus: overall level of adjustment Context: work focus, but items not specific for work contexts	Lounsbury et al. (2003) (6 items) Lounsbury et al. (2007)
Wanberg and Banas (2000)		19 items	Usage of validated scales	Rationale: personal resilience as personal characteristic Focus: (1) self-esteem; (2) perceived control; (3) optimism Context: no work focus	
Reivich and Shatté (2002)	Resilience Factor Inventory (RFI)	60 items	Construct (EFA; CFA) Criterion (PredV)	Rationale: resilience as malleable personal state Focus: (1) emotion regulation; (2) impulse control; (3) causal analysis; (4) self-efficacy; (5) realistic optimism; (6) empathy; (7) reaching out Context: specific for work contexts	Harker et al. (2016)

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Connor and Davidson (2003)	Connor-Davidson Resilience Scale (CD-RISC)	25 items	Construct (EFA; ConV; DisV)	Rationale: resilience as modifiable ability Focus: (1) personal competence, high standards, and tenacity; (2) trust in one's instincts, tolerance of negative affect, and strengthening effects of stress; (3) positive acceptance of change, and secure relationships; (4) control; (5) spiritual influences Context: no work focus	Gabriel et al. (2011) Guo et al. (2017) Hudgins (2016)
Sinclair and Wallston (2004) Ferris et al. (2005)	Brief Resilient Coping Scale (BRCS) Personal Resilience	4 items 6 items	Construct (EFA; ConV; DisV) Criterion (PredV) Not validated	Rationale: resilience as positive coping behaviour Focus: active coping and growth Context: no work focus Rationale: resilience as capacity for successful adaptation Focus: (1) nutrition; (2) physical activity; (3) leisure time; (4) personal relationships; (5) social networks; (6) commitment to change Context: no work focus	Bullough et al. (2014) Mache et al. (2014) Shoss et al. (2018)
Ferris et al. (2005)	Job Resilience	9 items	Not validated	Rationale: job resilience as the perception of job conditions Focus: (1) supervisor relationships; (2) peer relationships; (3) career opportunities; (4) company support; (5) rewards; (6) job definition; (7) physical environment; (8) decision making/control; (9) job security Context: specific for work contexts	

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Harland et al. (2005)		4 items	Construct (EFA)	Rationale: resilience as the degree to which a person grows and develops as a result of challenging experience Focus: learning and growth outcome orientation Context: no work focus	
Luthans et al. (2007b)	Resilience Scale [sub-scale of the PsyCap questionnaire; adapted from Wagnild and Young (1993)]	6 items	Validated in Luthans et al. (2007a) Construct (CFA; ConV; DisV) Criterion (PredV)	Rationale: resilience as maleable, positive psychological state Focus: handling challenges at work and recovery from them Context: specific for work contexts	Jung and Yoon (2015) (4 items) Martinez-Corts et al. (2015) (3 items) Verleysen et al. (2015) (3 items)
Oginska-Bulik and Juczynski (2008)	Resiliency Assessment Scale	25 items	Construct (CFA; ConV; DisV)	Rationale: resiliency as a personal trait, which promotes coping Focus: (1) determination and persistence in actions; (2) openness to new experiences and a sense of humour; (3) competencies to cope and tolerance of a negative affect; (4) tolerance of failures and treating life as a challenge; (5) optimistic life attitude and ability to mobilise in difficult situations Context: no work focus	Oginska-Bulik and Kobylarczyk (2015)

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Smith et al. (2008)	Brief Resilience Scale (BRS)	6 items	Construct (EFA, ConV) Criterion (PredV)	Rationale: resilience as a person ability to bounce back or recover from stress Focus: bouncing back from setbacks and recovery from them Context: no work focus	Crane and Searle (2016) Shoss et al. (2018)
Li et al. (2012)		6 items	Construct (CFA; DisV)	Rationale: emotional resilience as recovery from negative emotions Focus: quick recovery from negative and chaotic emotions Context: no work focus	
Lee et al. (2013)	Intrapersonal Resilience	58 items	Usage of validated scales	Rationale: intrapersonal resilience as personality trait Focus: (1) agreeableness; (2) conscientiousness; (3) extraversion; (4) emotional stability; (5) positive affect; (6) mastery Context: no work focus	
Lee et al. (2013)	Interpersonal Resilience [based on Sherbourne and Stewart (1991)]	19 items	Usage of validated scales	Rationale: interpersonal resilience as various forms of social support Focus: (1) affectionate support; (2) emotional/informational support; (3) positive social interaction; (4) tangible support Context: no work focus	

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
McLarnon and Rothstein (2013)	Workplace Resilience Inventory (WRI)	60 items	Construct (EFA; CFA) Criterion (PredV; IncV)	Rationale: workplace resilience as a skill that could be taught, practiced, and developed Focus: (1) initial responses; (2) affective personal characteristics; (3) behavioural personal characteristics; (4) cognitive personal characteristics; (5) opportunities, supports, and resources (6) affective self-regulatory processes; (7) behavioural self-regulatory processes; (8) cognitive self-regulatory processes	
Stephens et al. (2013)	[based on Caza and Bagozzi (2010)]	5 items	Construct (CFA)	Context: specific for work contexts Rationale: employee resilience as the extent to which they easily recover from negative events and regard those events as opportunities to grow and learn Focus: handling challenges at work and recover and grow from them	De Clercq and Belsaiteguioita (2017)
Winwood et al. (2013)	Resilience at Work Scale (RAW scale)	20 items	Construct (EFA; CFA); Criterion (PredV)	Context: specific for work contexts Rationale: workplace resilience as a skill that could be taught, practised, and developed Focus: (1) living authentically, (2) finding one's calling; (3) maintaining perspective; (4) managing stress; (5) interacting cooperatively; (6) staying healthy; (7) building networks Context: specific for work contexts	Malik and Garg (2017)

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Wei and Taormina (2014)		40 items	Construct (ConV) Criterion (PredV)	Rationale: resilience as personal quality Focus: (1) determination; (2) endurance; (3) adaptability; (4) recuperability Context: no work focus	
Näswall et al. (2015)	Employee Resilience Scale (EmpRes)	9 items	Construct (EFA)	Rationale: resilience as an adaptable employee capability, facilitated and supported by the organisation Focus: employee behaviour to utilise resources to continually adapt and flourish at work Context: specific for work contexts	Kuntz et al. (2017)
Mallak and Yildiz (2016)	Workplace Resilience Instrument (WRI)	20 items	Construct (EFA; CFA; ConV)	Rationale: workplace resilience as individual's ability to return to an original (or improved) condition after a stressful situation Focus: (1) active problem-solving; (2) team efficacy; (3) confident sense-making; (4) bricolage Context: specific for work contexts	
Meneghel et al. (2016a)		9 items	Construct (EFA; CFA)	Rationale: resilience as work-related ability Focus: bounce back, resist illness, adapt to stress, or thrive in the face of work-related adversity Context: specific for work contexts	
Braun et al. (2017)		6 items	Construct (EFA) Criterion (PredV)	Rationale: resilience as cognitive (e.g., framing), emotional, or behavioural adjustment to stress Focus: emotional and psychological transition related to change, responding effectively to either mitigate stress caused by the change, or manage or reduce increased stress Context: specific for work contexts	

TABLE 1
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Todt et al. (2018)	Innovator Resilience Potential (IRP)	18 items	Based on validated scales Construct (CFA; ConV; DisV) Criterion (IncrV)	Rationale: innovator resilience potential as innovators' predisposition to maintain their innovative performance after a setback like an innovation project termination Focus: (1) self-efficacy; (2) outcome expectancy; (3) optimism; (4) organisation-based self-esteem; (5) hope; (6) risk propensity Context: specific for innovation work contexts	

Construct Validation: EFA (exploratory factor analysis); CFA (confirmatory factor analysis); ConV (convergent validity); DisV (discriminant validity)
Criterion Validation: PredV (predictive validity); IncrV (incremental validity)
*Construct validation: We categorised validation procedures as construct validation in case authors showed that the measurement instrument was related (or not related) to variables/constructs that are known to be related (convergent validity) [or that are known to be not related (discriminant validity)] to the construct. Also, we categorised any forms of factor analysis as construct validation. The abbreviations are explained above.
**Criterion validation: We categorised validation procedures as criterion validation in case authors showed that the measurement construct was related to a predicted outcome or could explain more variance than existent measurement instruments of the same construct. The abbreviations are explained above.
***If a shortened scale was used (not the original), the number of items used in the shortened scale version is given in parentheses.

reflect manifestations of positive behaviour and adaptation despite adversity. Finally, to reflect a process understanding of resilience in the workplace, McLarnon and Rothstein (2013) developed a resilience scale that combines individuals' personal characteristics, availability of social support networks, initial responses to a significant traumatic event, and affective, cognitive, and behavioural self-regulatory processes. This aligns with the differentiation of the experience of adversity, resilience mechanisms, resilience-promoting factors, and resilience outcomes as parts of the resilience process.

From the 30 measurement instruments we identified, 13 instruments do not have a work focus, but investigate resilience as a personal, general phenomenon. The other measurement scales are specified for different work-related contexts, such as scales for career resilience (e.g., London, 1993; Noe, Noe, & Bachhuber, 1990), general work environments (e.g., Mallak & Yildiz, 2016; Winwood, Colon, & McEwen, 2013), or innovation contexts (Todt et al., 2018) and thus contain (at least some) work specific items. Scholars applied different approaches to validating these measurement instruments, but for a few of them, no validation was reported at all.

Even though several measurement scales reflect different resilience dimensions, most scales are unidimensional. Only a few scales are multi-dimensional (e.g., McLarnon & Rothstein, 2013; Todt et al., 2018). Further, although most studies measured resilience using self-report instruments developed for this purpose, a few studies operationalised resilience via the presence of a positive outcome associated with resilient individuals, such as higher job satisfaction, lower burnout scores, or higher business success (e.g., Yang & Danes, 2015; Zunz, 1998).

As our review highlights, there is no consensus yet among researchers on how to measure resilience at the individual level in the workplace. Numerous different scales have been used and developed, as highlighted in Table 1. This is problematic for advancement of the field and is likely to impede meta-analytic investigations to quantitatively consolidate findings from the extant literature. Most importantly, and as we outline in Table 1, the measures of resilience might not be directly comparable, as they rely on different conceptualisations of resilience or are either context specific or applicable to different work contexts.

To provide some guidance for selecting a resilience measure, first and foremost the choice of the measurement instrument should reflect the underlying conceptualisation of resilience as a trait, capacity, or process. Researchers should explicitly highlight their underlying conceptualisation of resilience, which has not been common practice in prior work. In addition, they should specify if they are studying general resilience that may influence workplace behaviour or if they are studying a work-domain specific conceptualisation of resilience. Finally, the usage of measurement scales building on the

conceptualisation of resilience as a capacity are suitable to pinpoint the potential of individuals to be resilient in the face of potential future adversity. To investigate resilience mechanisms, we suggest that measurement scales focusing on behaviours are most suitable since these reflect what individuals actually do to respond to adversity. To measure whether individuals have actually been resilient after an adverse event, pre-post measures of outcomes, such as performance or well-being, or the absence of problems like burnout, seem most appropriate.

Antecedents of Individual Resilience in the Workplace

Research on the antecedents of resilience at the individual level can be categorised in studies related to (1) personality traits and cultural value orientations, (2) personal resources, (3) personal attitudes and mindsets, (4) personal emotions, and (5) work demands and resources. Prior work on the antecedents of resilience at the individual level of analysis, especially that which examines the role of work demands and resources, personal resources, and personal emotions, has generally conceptualised resilience as a state-like capacity that is malleable.

Personality Traits and Cultural Value Orientations. Empirical research suggests that certain personality traits are positively linked to resilience (Förster & Ducheck, 2017). For instance, Wei and Taormina (2014) argued that people who worry a lot tend to think more thoroughly about how to overcome obstacles, which might have a positive effect on resilience. In line with these arguments, they found a significant positive correlation between worry and certain dimensions of their resilience construct. Moreover, Wei and Taormina (2014) and Lyons, Schweitzer, and Ng (2015) discovered a positive relationship between the personality trait of conscientiousness and both general resilience in the workplace and career resilience. They argued that this resulted from the fact that conscientious people are more hardworking and organised, and therefore better able to deal with adversity. Further, these scholars found future-orientation (Wei & Taormina, 2014), openness to experience, and emotional stability (Lyons et al., 2015) to be positively related to general resilience in the workplace and career resilience.

With regard to cultural value orientations, Welbourne, Gangadharan, and Sariol (2015) found that ethnicity moderated the relationship between workplace incivility and job satisfaction as well as between workplace incivility and burnout in such a way that employees with high horizontal collectivism values, which emphasise sociability (Triandis & Gelfand, 1998), tended to be more resilient to incivility in the workplace. In contrast, their colleagues with high horizontal individualism values, which emphasise self-reliance, tended to

be less resilient. Similarly, Wei and Taormina (2014) found that Chinese values, which emphasise relationships, are positively associated with resilience.

Personal Resources. Scholars found that personal resources, such as having expertise related to the job (Cameron & Brownie, 2010) or the ability to effectively manage work demands (Jensen, Trollope-Kumar, Waters, & Everson, 2008), positively relate to resilience. Other researchers have established a positive link between different personal resources, such as self-efficacy (Guo et al., 2017; Lyons et al., 2015) and business confidence (Yang & Danes, 2015), and resilience. They argued that self-efficacy/confidence acts as a personal resource to make individuals feel able to better address work challenges, in turn making them more persistent and motivated. Scholars have also found that an individual's feelings of control over a situation are positively related to his or her resilience. For example, based on qualitative interviews with physicians, Stevenson, Phillips, and Anderson (2011) found that the physicians' internal locus of control appeared to foster their resilience, and based on a quantitative study of managers and professionals, Lyons et al. (2015) found that an individual's external locus of control was negatively associated with his or her career resilience. Moreover, Kinman and Grant (2011) showed that the personal resource of reflective ability, comprised of the competencies of self-reflection and reflective communication, exhibited a positive relationship with resilience. Finally, scholars have suggested that the ability to establish balance by setting limits and through preserving adequate work-life balance fosters resilience, as rest and social contact during leisure time offer the opportunity to refill depleted resources (Jensen et al., 2008; Meek et al., 2003). Even though the findings of these studies are consistent with theories including the conservation of resources theory (Hobfoll, 1989) and the job demands-resources theory (Demerouti, Bakker, Nachreiner, & Schaufeli, 2001), scholars did not make use of those theories in explaining their findings.

A strong link has been found between employees' social competencies, such as emotional intelligence or empathy, and their resilience (Förster & Ducheck, 2017; Kinman & Grant, 2011). Referring to the theory of high-quality connections (Dutton & Heaphy, 2003), which states that connections at work are vital for employees' health and well-being, Stephens, Heaphy, Carmeli, Spreitzer, and Dutton (2013) found that an individual's emotional carrying capacity, defined as the capacity to constructively express positive and negative emotions within a relationship, was positively related to individual resilience and mediated the effects of relationship closeness on resilience. Consistent with self-determination theory (Deci & Ryan, 2000), researchers have also found that when employees' needs for competence are satisfied, they tend

to exhibit higher levels of resilience (Verleysen, Van Acker, & Lambrechts, 2015).

Personal Attitudes and Mindsets. There is burgeoning evidence that finding a sense of purpose and meaning (e.g., Cameron & Brownie, 2010; Stevenson et al., 2011), and maintaining one's sense of professional mission (Zunz, 1998), help employees to become resilient in times of adversity. This is because a sense of meaning can offer orientation in an unstable environment. For example, in exploring the resilience paths of entrepreneurs who perceived their ventures to have failed, Corner et al. (2017) found that having other responsibilities and commitments was related to entrepreneurs' perceptions of purpose and to limited disruptions in emotional and psychological functioning. Moreover, in caring professions, a sense of calling and spirituality seems to foster employees' resilience (Ablett & Jones, 2007; Meek et al., 2003).

Several studies have investigated how personal attitudes towards one's job and career relate to resilience. For instance, Ablett and Jones (2007) found that being committed to one's job positively relates to resilience. Further, Lyons et al. (2015) found that being self-directed fostered career resilience, whereas being values-driven, having a boundaryless mindset, and having a preference for organisational mobility seemed to diminish career resilience.

Personal Emotions. Other researchers have begun to examine the influence of emotions on resilience. In doing so, they conceptualise resilience as a state-like developable capacity that results from the experience of positive emotions at work. Sommer, Howell, and Hadley (2016) found that positive emotions fostered employees' resilience and negative emotions diminished employees' resilience during an organisational crisis. The scholars explain their findings with the broaden-and-build theory (Fredrickson, 2001), which suggests that the experience of positive emotions broadens people's momentary thought-action repertoire and in turn leads to the experience of a more positive affect and emotional well-being. Although other studies have not explicitly drawn on the broaden-and-build theory, most of them have been supportive of such a perspective. For example, Cameron and Brownie (2010) highlighted that maintaining a positive attitude through the use of humour fostered resilience. Adding to the empirical evidence, Green, Schaefer, MacDermid, and Weiss (2011) found that negative emotional displays by an employee's partner were negatively related to career resilience.

Work Resources and Demands. Beyond personal and interpersonal factors, researchers have begun to explore how the work context in which employees are situated is related to their resilience. In this regard, scholars

have highlighted how resources from the work context promote employees' resilience. This is in line with the conservation of resources theory (Hobfoll, 1989), which treats resilience as a personal resource that may be influenced by social resources from the contextual environment. Surprisingly, only a few studies referred to this theory when explaining research ideas and findings. With regard to work resources, researchers found social support and feedback from others in the workplace to positively relate to resilience (e.g., Förster & Duchek, 2017; Kuntz, Connell, & Näswall, 2017; McDonald, Jackson, Vickers, & Wilkes, 2016). Employees benefitted from sharing responsibilities and work tasks with colleagues, as this relieved the burden in challenging situations (Burns, Poikkeus, & Aro, 2013). Social support also provided individuals with the opportunity to talk about and work through stressful experiences (Lamb & Cogan, 2016) and to debrief after experiencing challenges at work (Cameron & Brownie, 2010). Todt et al. (2018) found that, whereas the social support provided by colleagues and supervisors was strongly related to employees' resilience, social support from family and friends was not. In contrast, Jensen et al. (2008) found that strong professional and private social networks promoted resilience. These conflicting findings may result from the different work contexts studied (i.e., innovation context and healthcare).

Also looking at the role of social support at work, Kuntz, Connell, and Näswall (2017) found that the positive effect of colleagues' and supervisors' social support as well as performance feedback from supervisors on employees' resilience was stronger for employees with a high promotion focus in combination with a high prevention focus. Similarly, they found that a high promotion in combination with a high prevention focus diminished the negative effect of low feedback availability (Kuntz, Connell, & Näswall, 2017). Other researchers have examined the role played by the supervisor in promoting resilience (e.g., Peters & Pearce, 2012). For example, Noe et al. (1990) found that the level of support provided by a manager was positively related to career resilience. In contrast, Kidd and Smewing (2001) did not find a relationship between supervisor support and career resilience. Notably, the two studies used different scales to measure career resilience, which might explain the conflicting results.

Researchers have also begun to examine the link between leadership styles and employees' resilience. For example, Sommer et al. (2016) reported that transformational leadership was positively associated with greater levels of positive affect and lower levels of negative affect in subordinates, subsequently fostering resilience among team members. Harland, Harrison, Jones, and Reiter-Palmon (2005) showed that the transformational leadership dimensions of attributed charisma, idealised influence, intellectual stimulation, and individualised consideration, as well as the transactional leadership

dimension of contingent reward, were positively related to subordinates' resilience. They found a negative relationship between the passive form of management-by-exception and subordinates' resilience and no relationship between the active form of management-by-exception and resilience.

Finally, researchers have investigated the effects of organisational contexts. Meneghel, Borgogni, Miraglia, Salanova, and Martínez (2016a) found that employees' perceptions of a positive social context were positively linked to their resilience. They drew on conservation of resource theory (Hobfoll, 1989) to explain their findings. According to this theory, access to contextual resources from the social environment provides fertile ground for the development of personal resources, such as resilience. In line with the broaden-and-build theory (Fredrickson, 2001), Malik and Garg (2017) further found that learning culture, inquiry and dialogue, and knowledge sharing structure are positively related to employees' resilience. Specifically, they argued that the experience of positive organisational environments fosters positive emotions, which leads to the development of resilience.

Outcomes of Individual Resilience

We categorise research on outcomes of resilience at the individual level into four categories: (1) performance, (2) mental and physical health, (3) work-related attitudes, and (4) change-related attitudes. Prior work on the outcomes of resilience has tended to conceptualise resilience as a stable personality trait or developable state-like capacity that enables individuals to perform at higher levels, deal with stressful events at work, and adopt a more positive mindset in the workplace.

Performance. Prior research has shown that an employee's resilience positively relates to individual job performance, both when self-rated and when supervisor-rated (Luthans, Avolio, Walumbwa, & Li, 2005; Luthans, Avolio, Avey, & Norman, 2007a). Jung and Yoon (2015) also reported that resilient individuals were more likely to engage in organisational citizenship behaviour. Such work conceptualises resilience as a trait or personal state-like capacity that enables employees to maintain high levels of motivation and exert more effort at work. Further, resilience has been found to positively relate to employees' own evaluations of their career success (Wei & Taormina, 2014).

Mental and Physical Health. Researchers have also examined the link between resilience and different measures of mental and physical health. These researchers have argued that resilience should be treated as a personality trait or state-like capacity that buffers the employee from

demands or negative experiences at work. This argument is in line with resource theories such as the job demands-resources theory (Demerouti et al., 2001), yet studies have not linked their findings to this theory. Scholars have found a positive relationship between employees' resilience and their mental health (Kinman & Grant, 2011), a negative relationship with symptoms of burnout and emotional exhaustion (Cooke, Doust, & Steele, 2013; Harker, Pidgeon, Klaassen, & King, 2016; Shoss, Jiang, & Probst, 2018), and a positive relationship with post-traumatic growth (Ogińska-Bulik & Kobylarczyk, 2015). McLarnon and Rothstein (2013) found that four of the eight dimensions of their workplace resilience inventory were negatively correlated with symptoms of depression. Further, Ferris, Sinclair, and Kline (2005) found that the perceived lack of both job resilience, defined as the perception of negative work conditions, and personal resilience predicted biopsychosocial strain, which encompassed sleep problems, poor emotional health, low energy, limited concentration, and high levels of perceived stress. High biopsychosocial strain, in turn, predicted a high risk of cardiovascular disease. In examining these issues, the authors theoretically linked their findings to the job demand-control-support model (Johnson & Hall, 1988) and the effort-reward imbalance model (Siegrist, Peter, Junge, Cremer, & Seidel, 1990). Parker, Jimmieson, Walsh, and Loakes (2015) found that resilience tended to enhance coping abilities, especially if employees had high levels of control over their work, as this seemed to be preferred by resilient individuals.

Work-Related Attitudes. There is growing evidence that resilience is positively related to employees' satisfaction in both their jobs (e.g., Badran & Youssef-Morgan, 2015; Youssef & Luthans, 2007) and their careers (e.g., Lounsbury et al., 2003; Lyons et al., 2015). In addition, resilience has been linked with increased work happiness (Youssef & Luthans, 2007) and reduced cynicism at work (Shoss et al., 2018). Also, scholars found a positive link between resilience and work-related attitudes that are desirable from an employer's and an employee's perspective. For example, Youssef and Luthans (2007) and Larson and Luthans (2006) demonstrated a strong link between resilience and organisational commitment, and Mache et al. (2014) observed a positive relationship between resilience and work engagement. Finally, Cho, Park, and Dahlggaard-Park (2017) found resilience to positively relate to an employee's perception of his/her psychological contract with the employer. Whereas employees' resilience was positively associated with their perceptions of relational contracts, it was negatively related to their perceptions of transactional psychological contracts with their employers (Cho et al., 2017).

Change-Related Attitudes. Wanberg and Banas (2000) found a positive association between resilience and employees' openness to organisational change. Shin et al. (2012) showed that resilience was positively related to employees' normative and affective commitment to change via driving positive affect. Confirming this finding, Malik and Garg (2017) found that resilience positively relates to affective commitment to change. Shin et al. (2012) drew on the conservation of resources theory (Hobfoll, 1989) to propose that individuals' personal resources, such as resilience, can increase their coping abilities by providing energy or protecting against dysfunctional psychological states triggered by a stressor such as change. Gowan, Craft, and Zimmermann (2000) found that career-resilient army staff anticipated fewer negative consequences when making the transition to civilian life. They explained their findings with the model of stress, appraisal, and coping (Lazarus & Folkman, 1984), which states that individuals' interpretations of stressful events depend upon their perception of available resources to deal with the consequences of those events. Complementing this, Carless and Bernath (2007) found that career-resilient psychologists tended to have a lower intent to change careers and Kidd and Green (2006) found that career resilience lowered biomedical research scientists' intention to leave science. Finally, scholars found resilience to positively relate to an individual's intent to start a business (Bullough, Renko, & Myatt, 2014).

Individual Resilience as a Mediator

Conceptualising resilience as a state-like developable capacity, pioneering studies have started to investigate the mediating effects of resilience at the individual level. Based on the challenge-hindrance stressors framework (Cavanaugh, Boswell, Roehling, & Boudreau, 2000), which states that challenging stressors are potentially rewarding and that hindering stressors are inhibiting, Crane and Searle (2016) found resilience to fully mediate both the negative relationship between challenge stressors and strain and the positive effect of hindrance stressors on strain. Similarly, Kinman and Grant (2011) found resilience to fully mediate the negative relationship between emotional intelligence and mental distress. Malik and Garg (2017) found resilience to partially mediate the positive relationship of learning culture, inquiry and dialogue, and knowledge sharing structure with affective commitment to change. Finally, Lyons et al. (2015) found that career resilience partially mediated the positive relationships of emotional stability, career self-evaluation, and modern career orientation with career satisfaction.

To summarise the reviewed research so far, Figure 2 provides an overview and categorisation of the empirical work on antecedents and outcomes of individual resilience in the workplace and individual resilience as a mediator.

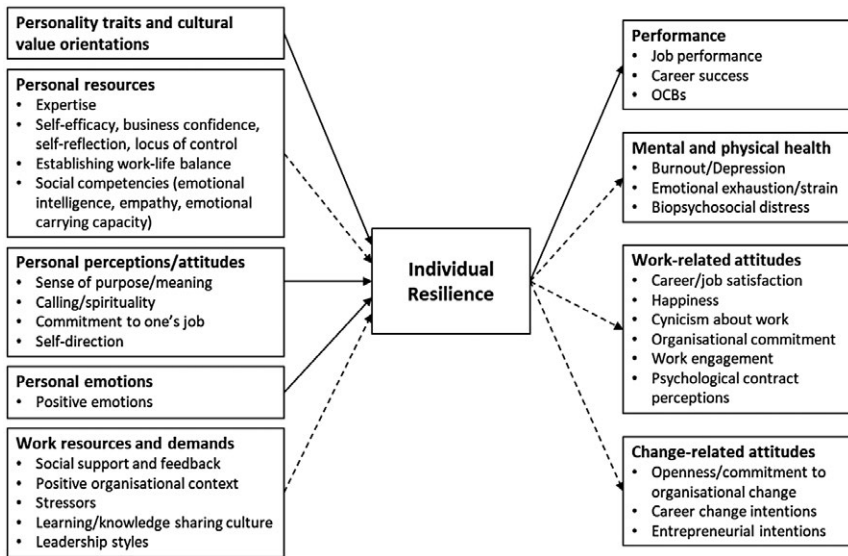


FIGURE 2. Overview and categorisation of existing research on individual resilience in the workplace: Antecedents and outcomes of individual resilience and individual resilience as mediator.

Note: Where research has investigated direct and mediated relationships, we have used dashed arrows, and where research has only investigated direct relationships (including qualitative studies), we have used solid arrows.

Individual Resilience as a Moderator

Scholars have also begun to examine whether resilience reduces the harmful influence of job demands or negative experiences at work. In doing so, some have drawn on the job demands-resources theory (Demerouti et al., 2001) to argue that resilience functions as a personal resource that buffers employees from the unfavourable side effects of job demands like conflict at work. Figure 3 summarises this empirical work and provides an overview and categorisation of the empirical research that has investigated individual resilience as a moderator.

For example, De Clercq and Belausteguigoitia (2017) found that resilience buffered employees from the negative effects of organisational politics at work on organisational citizenship behaviour. They also found that this buffering effect was stronger at lower levels of transformational leadership. Further, Shoss et al. (2018) found that resilience buffered against the negative effects of job insecurity on emotional exhaustion, cynicism, and perceptions of psychological contracts. In a daily diary study with nurses, Gabriel

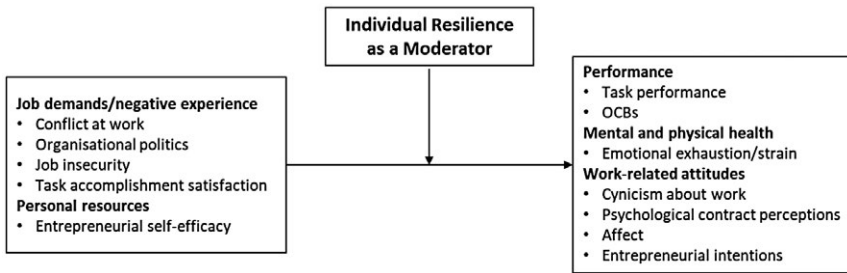


FIGURE 3. Overview and categorisation of existing research on individual resilience in the workplace: Individual resilience as moderator.

et al. (2011) found that resilience moderated the within-person relationship between indirect care task accomplishment satisfaction and post-shift positive affect. Those individuals high in resilience had high positive affect regardless of the level of task accomplishment satisfaction. Also conducting a daily diary study, Martinez-Corts, Demerouti, Bakker, and Boz (2015), found that resilience prevented the spillover of interpersonal work conflict into the non-work domain. On days when employees experienced lower levels of resilience, strain resulting from interpersonal work conflict decreased their ability to handle non-work demands. Similarly, van Erp, Rispen, Gevers, and Demerouti (2015) found that resilience alleviated the negative effects of bystander conflicts, that is, conflicts that arise if employees are prevented from pursuing their work by persons who are not involved in the process. Such situations are typical in emergency settings, such as accidents. In contrast to other studies that examined the buffering effects of resilience, suggesting it is a loss-oriented personal resource, Bullough et al. (2014) found that the positive relationship between entrepreneurial self-efficacy and entrepreneurial intentions was stronger for individuals with high levels of resilience, suggesting it might also act as a gain-oriented personal resource.

TEAM RESILIENCE IN THE WORKPLACE

Conceptualising Resilience at the Team Level

In recent years, a number of scholars have started to elaborate on resilience as a collective phenomenon, which is conceptually different from individual resilience (e.g., Carmeli et al., 2013). Yet, the conceptual development of team resilience is still in its infancy, as is comprehensively summarised by Bowers, Kreutzer, Cannon-Bowers, and Lamb (2017) and Morgan, Fletcher, and Sarkar (2017). Given the nascent stage of theorising on team resilience,

no consensus has yet emerged with regard to conceptualising this team level phenomenon (Kennedy, Landon, & Maynard, 2016). For example, West et al. (2009) argue that team resilience can be regarded as an isomorphic representation of the individual resilience capacities. Such an understanding implies that team resilience is a shared construct, which emerges through composition (Kozlowski & Klein, 2000). Stoverink et al. (2018) conceptualise team resilience as an emergent state, which “characterizes properties of the team that are typically dynamic in nature and vary as a function of team context, inputs, processes, and outcomes” (Marks, Mathieu, & Zaccaro, 2001, p. 357). This perspective highlights the team level of analysis and describes team resilience as an outcome of contextual factors and team member interactions. Research on other collective-level constructs, such as collective efficacy or collective creativity, has highlighted the importance of explicitly accounting for coordinative, relational, and interactive aspects of teams (Bandura, 2000; Taggar, 2002). We advocate for conceptualising team resilience as an emergent process that originates through these interactions but manifests as higher-level phenomena. An outcome of this emergent process could be a team resilient state, which is why an understanding of team resilience as an emergent process does not exclude an understanding of team resilience as a team emergent state. Kozlowski and Klein (2000) differentiate between shared and configural emergent processes. In their pure form, shared processes describe the convergence of lower-level characteristics through which a higher-level property emerges. In contrast, configural processes describe a more complex grouping of diverse lower-level contributions that together form a higher-level property. We suggest that resilient team members may not necessarily form a resilient team, which would be in line with the latter perspective. However, team members that collaborate over a longer time may develop shared perceptions of their team’s resilience, rather reflecting the former perspective. Kozlowski and Klein (2000) note that the type of an emergent process may vary for a given phenomenon depending on the nature of the social-psychological interactions. Thus, in future research, scholars may want to investigate in which cases team resilience may be best understood as either a shared emergent process or a configural emergent process.

Measuring Team Resilience

In empirical quantitative research on team resilience, scholars have conceptualised resilience as a team capacity or ability. Table 2 provides an overview of the six different self-report measurement instruments that have been used or developed in the articles on team resilience in the workplace

TABLE 2
Overview on Measurement Instruments Developed/Used to Measure Team Resilience in the Workplace

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Block and Kremen (1996)	Ego-Resiliency Scale (ER89)	14 items	Validated for individual level: Construct (ConV; DisV)	Rationale: resilience as a generalised, characterological individual quality Focus: ability to change from and also return to the individual's characteristics level of ego-control	Yang et al., (2015) (5 items)
Luthans et al. (2007b)	Resilience Scale [sub-scale of the PsyCap questionnaire; adapted from Wagnild and Young (1993)]	6 items	Validated for individual level in Luthans et al. (2007a)	Context: no work focus Rationale: resilience as malleable, positive psychological state Focus: handling challenges at work and recovery from them Context: specific for the work context	West et al. (2009) [with referent shift]
Salanova et al. (2012)	Team Resilience [subscale of Healthy and Resilient Organization (HERO) Scale]	7 items	Validation of overarching HERO Scale	Rationale: team resilience as a team capacity Focus: attitudes and behaviours [based on Mallak (1998)] Context: specific for the work context	Meneghel et al. (2016c) Meneghel, Martínez, et al. (2016b)
Carmeli et al. (2013)		6 items	Construct (EFA) Based on validated scales	Rationale: team resilience as team's beliefs and capacities Focus: (1) team-efficacious beliefs; (2) team-adaptive capacity Context: specific for the work context	

TABLE 2
Continued

<i>Authors</i>	<i>Scale Name</i>	<i># of Items</i>	<i>Validation***</i>	<i>Underlying Conceptualisation and Focus</i>	<i>Examples of Usage in Research on Resilience in the Workplace***</i>
Stephens et al. (2013)		3 items	Construct (EFA)	Rationale: team resilience as a team capacity Focus: coping with difficult situations Context: specific for the work context	
Oeij (2017)	Team Innovation Resilience Behavior Scale (IRB) [based on the Safety Organizing Scale by Vogus and Sutcliffe (2007)]	18 items	Based on a validated scale	Rationale: team resilience as a team behaviour Focus: (1) preoccupation with failure; (2) reluctance to simplify; (3) sensitivity to operations; (4) commitment to resilience; (5) deference to expertise Context: specific for the work context	Oeij et al. (2017)

Construct Validation: EFA (exploratory factor analysis); CFA (confirmatory factor analysis); ConV (convergent validity); DisV (discriminant validity)
Criterion Validation: PredV (predictive validity); IncrV (incremental validity)
*Construct validation: We categorised validation procedures as construct validation in case authors showed that the measurement instrument was related (or not related) to variables/constructs that are known to be related (convergent validity) [or that are known to be not related (discriminant validity)] to the construct. Also, we categorised any forms of factor analysis as construct validation. The abbreviations are explained above.
**Criterion validation: We categorised validation procedures as criterion validation in case authors showed that the measurement construct was related to a predicted outcome or could explain more variance than existent measurement instruments of the same construct. The abbreviations are explained above.
***If a shortened scale was used (not the original), the number of items used in the shortened scale version is given in parentheses.

we reviewed. In five of these empirical studies, team resilience was operationalised as a shared team capacity (Carmeli et al., 2013; Meneghel et al., 2016b; Meneghel, Salanova, & Martínez, 2016c; Oeij, Dhondt, Gaspersz, & van Vuuren, 2017; Stephens et al., 2013). Carmeli et al. (2013) and Stephens et al. (2013) designed new team resilience measurement scales. In contrast, Meneghel et al. (2016b, 2016c) and West et al. (2009) made use of existing scales developed for investigating resilience of organisations and individuals. In each of these studies, team members' perceptions of the team's resilience were aggregated to the team level based on high rater agreement (r_{wg}) or high intra-class correlation coefficients (ICCs) between team members, which demonstrate the shared nature of the team-level constructs (Bliese, 2000). In contrast to these studies, and using Block and Kremen's (1996) ego-resilience scale, Yang, Lee, and Cheng (2015) investigated team resilience as the mean of the individual resilience of team members. The authors aggregated resilience at the individual level to reflect the team level. Finally, Oeij et al. (2017) relied on a measurement instrument that Oeij (2017) had developed for teams based on Vogus and Sutcliffe (2007).

Our review suggests that more work is needed to establish a well-validated measurement instrument for team resilience. As team resilience is conceptually different from resilience at the individual level, we do not recommend measuring team resilience as the mean of team members' individual resilience. Such a measure is unlikely to fully capture the social nature of team resilience. Rather, measurement scales should reflect the nature of a team construct (Kozlowski & Klein, 2000). Except for one study (Stephens et al., 2013), all measurement instruments for the team level were based on validated scales. In two cases, the measurement instruments were validated on the basis of exploratory factor analysis (Carmeli et al., 2013; Stephens et al., 2013). Even though measurement scales mostly relied on validated scales, these have not necessarily been validated for the team level of analysis. Thus, we suggest that more work that validates these scales for application within the context of work teams could be valuable.

Antecedents of Team Resilience in the Workplace

Empirical research on resilience has also started to examine the antecedents of resilience at the team level. We categorise existing research on antecedents of resilience at the team level into three main groups: (1) emotions, (2) interpersonal processes, and (3) structural aspects.

Emotions. Meneghel et al. (2016c) found that collective positive emotions like shared enthusiasm, optimism, comfort, or relaxation tend to enhance resilience at the team level. Drawing on the broaden-and-build

theory (Fredrickson, 2001), they argued that the collective experience of positive emotions enhances employees' access to important team resources, which in turn fosters a team's resilience (Meneghel et al., 2016c).

Interpersonal Processes. With regard to interpersonal processes, scholars have primarily focused on relational aspects. Carmeli et al. (2013) found that team connectivity as a structural tie had a positive direct relationship with team resilience. It also had a positive indirect relationship with team resilience through enhancing strategic decision comprehensiveness, defined as the systematic collection and processing of information. Drawing on the broaden-and-build theory (Fredrickson, 2001) and the theory of high-quality connections (Dutton & Heaphy, 2003), Carmeli et al. (2013) argued that the nature of relationships has important implications for the experience of positive emotions and facilitates cognitive processes. Complementing these results, Stephens et al. (2013) found that emotional carrying capacity in teams was positively related to team resilience and mediated the effects of intra-team trust on team resilience. Drawing on the theory of high-quality connections, they argued that emotional expression between team members fosters team solidarity (Stephens et al., 2013). Moreover, the sharing of negative emotions appeared to help teams work through team members' concerns, while the sharing of positive emotions contributed to recovering from these concerns (Stephens et al., 2013). Similarly, Meneghel et al. (2016b) found that social job resources, in the form of a social support climate and team coordination, positively related to team resilience. This effect was moderated by job demands in such a way that the impact of job resources on team resilience was weakened in the case of high job demands. To explain these effects, they referred to the conservation of resources theory (Hobfoll, 1989), which suggests that teams strive to accumulate resources. Therefore, resource-rich environments are likely to foster additional personal resources, such as resilience.

Another interpersonal process that has been examined in the literature to explain resilience at the team level is communication within the team. Based on the analysis of an emergency room team that underwent a nuclear disaster simulation, Gomes, Borges, Huber, and Carvalho (2014) found that through engaging in briefing and debriefing activities, the team built a shared situation awareness of the emergency event. This helped in coordinating collective actions that enabled a resilient response to the situation.

Structural Aspects. Stuart and Moore (2017) investigated how the loss of certain team roles affected team resilience. They found that the loss of a team member that occupied an illicit organisational role was negatively

related to teams' performance, especially if the role owner was a long-term member of the team. Further, they found that a team's performance tended to recover more slowly when the team tried to replace the loss of a team member in an illicit role.

Outcomes of Resilience at the Team Level

Team Performance. Meneghel and colleagues (2016b, 2016c) observed that team resilience had a positive relationship with in-role and extra-role team performance. They argued that this was because resilient teams are better able to find flexible and effective solutions when confronted with challenges and adversity.

Team Attitudes and Behaviours. In their study on newly formed project teams, West et al. (2009) found that team resilience was positively associated with team cohesion and cooperation, but only in later project phases, as these qualities took time to develop. Their rationale was that teams capable of managing challenges were more likely to stay supportive and be willing to communicate.

Team Resilience as a Mediator

In their studies of team resilience, Meneghel et al. (2016b, 2016c) found that team resilience fully mediated the effects of collective positive emotions on in-role and extra-role team performance (Meneghel et al., 2016c) and partially mediated the effects of social support on in-role and extra-role team performance (Meneghel et al., 2016b).

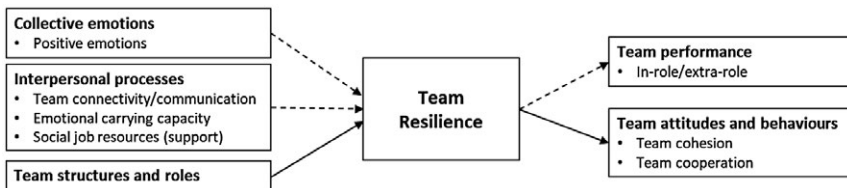


FIGURE 4. Overview and categorisation of existing research on team resilience in the workplace: Antecedents and outcomes of team resilience and team resilience as mediator.

Note: Where research has investigated direct and mediated relationships, we have used a dashed arrow, and where research has only investigated direct relationships (including qualitative studies), we have used a solid arrow.

To summarise the research reviewed so far, Figure 4 provides an overview and categorisation of the empirical work on antecedents and outcomes of team resilience in the workplace and team resilience as a mediator.

Team Resilience as a Moderator

We found only one study that concentrated on team resilience as a moderator. Drawing on the conservation of resources theory (Hobfoll, 1989), Yang et al. (2015) investigated the moderating effect of team members' mean level of resilience. They found that the resilience of frontline teams in service firms positively moderated the relationship between operational improvement competence and service recovery performance and weakened the negative effect of role conflict on this relationship.

IMPLICATIONS AND AGENDA FOR FUTURE RESEARCH

Opportunities for Theoretical Advancement

Based on our review, we found that there is a lack of theory-driven empirical resilience research. Existing empirical research has provided only a limited understanding of the specific processes by which resilience develops and influences positive and negative work outcomes, or the boundary conditions that affect these relationships. Accordingly, we call for elaborated theoretical foundations of research on resilience in the workplace. In the following sections, we highlight potential theoretical frameworks that researchers may draw on to examine these issues. Pioneering research has started to draw on the job demands-resources theory (Demerouti et al., 2001), the conservation of resources theory (Hobfoll, 1989), and the broaden-and-build theory (Fredrickson, 2001). Trait-activation theory (Tett & Guterman, 2000) and social cognitive theory (Bandura, 1986) offer complementing theoretical lenses that may be applied to better understand resilience in the workplace.

Integrated Resource Theories. In light of work viewing resilience as a personal resource that individuals draw upon to maintain psychological well-being, researchers could consider drawing on integrated resource theories such as the job demands-resources theory (Demerouti et al., 2001) and conservation of resources theory (Hobfoll, 1989) to explain how resilience develops and influences outcomes. For example, researchers might consider drawing on the job demands-resources model to investigate whether resilience is a personal resource that acts as a buffer against the negative influence of work demands in the workplace. For instance, De Clercq and Belausteguigoitia (2017), Martinez-Corts et al. (2015), and Gabriel et al. (2011) began to examine the role played by resilience in buffering the negative influence of work demands.

Further research may build on this work to better understand resilience as a personal resource. Also, scholars might draw on the key tenets of the conservation of resources theory, which distinguishes between loss-oriented personal resources that protect against resource loss and gain-oriented personal resources that help individuals acquire further resources (Hobfoll, 1989). Pioneering research (Meneghel et al., 2016b; Shin et al., 2012) has drawn on the conservation of resources theory (Hobfoll, 1989) and highlighted a gain-oriented perspective to explain how resilience can help individuals obtain additional resources from the environment.

Trait-Activation Theory. As highlighted in our review, several studies on resilience at the individual level have conceptualised resilience as a stable or malleable personal characteristic. To better understand the boundary conditions of the relationship between resilience and its outcomes, we suggest integrating person-situation theoretical perspectives, such as the trait-activation theory (Tett & Guterman, 2000). According to the trait-activation theory, the behavioural expression of personality traits is dependent on the arousal of these traits by trait-relevant situational cues (Tett & Guterman, 2000). Thus, situational cues, such as organisational or team contexts, may influence how resilience as a personal characteristic is related to outcomes of interest, such as work behaviours and attitudes. Whether the relationship is influenced by the situation depends on whether the situation is trait-relevant and fosters trait expression (Tett & Guterman, 2000). As such, trait-activation theory may help explain how organisational and team environments foster individuals' resilience. Based on the trait-activation theory, we argue that supportive environments and teams with high levels of psychological safety or collective humility are relevant to the expression of resilience and amplify the positive effects of an individual's resilience. For example, such environments may support employees in addressing the adversity they face by providing informational or emotional support from colleagues.

Social Cognitive Theory. As prior work on resilience has often conceptualised resilience as a trait or state-like capacity, we encourage scholars to investigate whether resilience is a developable process. As such, employees and teams may build their resilience through leveraging their network and through adaptive behaviours (Kuntz, Malinen, & Näswall, 2017). Also, contingencies may play a role in forming resilience (Moenkemeyer et al., 2012). In conceptualising resilience as a developable process, researchers might draw on Bandura's (1986) social cognitive theory. According to social cognitive theory, psychosocial functioning can be explained in terms of triadic reciprocal causation (Bandura, 1986), that is, "behavior, cognitive and other personal factors, and environmental events all operate as interacting

determinants that influence each other bi-directionally” (Bandura, 1988, p. 276). According to this perspective, employees are producers of their personality, behaviour, and environment, yet at the same time are products of it. Applying this theoretical perspective to research on resilience in the workplace may help us better understand the dynamic nature of resilience (Moenkemeyer et al., 2012). Based on this perspective, scholars could investigate how an individual’s or team’s experience of adversity might influence their resilience over time (Almeida, 2005). Further, social cognitive theory may help illustrate how the development of resilience from one episode or event influences an individual’s or team’s response to future adversity. For instance, the successful handling of one adverse episode might strengthen efficacy and create positive emotions, enhancing an individual’s potential to successfully handle upcoming challenges (Fredrickson, 2001). Yet, as Luthans, Youssef, and Avolio (2007b) argued with regard to psychological capital, an inverted effect might also be possible, that is, if an individual (or team) is not resilient in facing adversity, this may weaken his/her (or its) ability to deal with subsequent adversity and result in a downward spiral. Hence, it is important to consider these dynamic and temporal effects.

Broaden-and-Build Theory. Researchers may consider utilising broaden-and-build theory (Fredrickson, 2001), which is a prominent theory in positive psychology that explains how resilience develops in general populations. According to broaden-and-build theory, positive emotions broaden people’s momentary thought-action repertoires and expand the range of available cognitions and behaviours. Over time, the experience of positive emotions can build positive resources, which helps individuals successfully cope with negative experiences and fosters resilience (Cohn, Fredrickson, Brown, Mikels, & Conway, 2009). Our review of literature on resilience in the workplace revealed that even though studies are generally supportive of this theoretical perspective, only a limited number of studies explicitly referred to the broaden-and-build theory (e.g., Meneghel et al., 2016c; Sommer et al., 2016). However, the theory is applicable at both the individual and the team levels of analysis (Fredrickson, 2003) and might thus have the potential to explain how team resilience develops from individual resilience in future multilevel work. Empirical evidence shows that positive emotional states help individuals recover from work-related stress (van Hooff & de Pater, 2017), which suggests that the theory might be applicable not only for general populations, but also for the work-context. We call on scholars to consider integrating this theoretical perspective into future research on resilience in the workplace.

Opportunities for Empirical Advancement

Conceptual Foundations of Resilience Research. Our systematic review of the literature uncovered that there are still considerable inconsistencies regarding how resilience has been conceptualised in management studies, which is in line with the finding of Fisher et al. (2018). Further, Fisher et al. (2018) noted that acknowledging temporal issues and the role of adversity is important in future empirical research on resilience in the workplace. Taking the rationale of this conceptual work further, we advocate a process conceptualisation of resilience (Fisher et al., 2018; Luthar et al., 2000; Moenkemeyer et al., 2012), which encompasses other conceptualisations of resilience (e.g., trait or capacity approaches) and allows for the study of temporal issues. An empirical challenge in research on resilience is that resilience encompasses (1) the actual experience of adversity, (2) the mechanisms and promoting factors individuals and teams draw on to deal with adversity, and (3) the manifestation of positive outcomes. Yet, in this regard, prior research has noted that it is important to differentiate between resilience as a capacity, describing the potential to be resilient in the face of adversity, and the demonstration of resilience after having faced adversity (Britt et al., 2016; Stoverink et al., 2018). In order to study the process that can foster the demonstration of resilience over time, we suggest that qualitative research approaches may be well suited, as qualitative research allows for the investigation of complex, processual dynamics and human interaction (Gioia, Corley, & Hamilton, 2013).

With regard to quantitative studies, those using a capacity perspective have mostly not documented the manifestations of adversity or positive adaptation. Rather, these studies have investigated the mere potential of individuals and teams to positively adapt in cases of adversity and have pinpointed the level of this resilience potential at a specific point in time. Yet, this research can also be informative and valuable in providing a better understanding of resilience in the workplace. Investigating and measuring resilience as a capacity, or as an emergent state at the team level, through quantitative research may help researchers identify different stages in the resilience process. We therefore call on researchers to be specific with regard to what aspect of the resilience process they are investigating and measuring (adversity, the potential for different entities to be resilient, resilience promoting factors and mechanisms, and/or a positive outcome).

Multi-Level Research on Resilience in the Workplace. In line with a general multi-level perspective on organisational phenomena (Kozlowski & Klein, 2000), we suggest that individual resilience and team resilience are interdependent and can mutually influence each other. Thus, we encourage adoption of a multi-level lens to examine resilience. As our review highlights,

scholars have tended to focus on one level of analysis when investigating the antecedents and outcomes of resilience. Yet, our review shows that interpersonal relationships and positive emotions may play an important role in fostering resilience at the individual and team levels. Investigation of these topics may provide a starting point to connect the two levels of analysis once issues of how to measure resilience appropriately at the different levels are sorted out (Kozlowski & Klein, 2000). For example, researchers might draw on the broaden-and-build theory (Fredrickson, 2001) to investigate whether individual resilience fosters team resilience. According to this theory, the display of positive emotions can become an automatic behaviour for resilient individuals (Fredrickson, 2001). Through momentary affective transfer processes, positive emotions can spread from resilient individuals to other members of a team, leading to the emergence of collective positive emotions (Barsade & Knight, 2015). For instance, through a process of emotional contagion, individuals may start to mimic the facial expressions of team colleagues, leading to similar affective experiences (Hatfield, Rapson, & Cacioppo, 1994). Thus, individual resilience may have a positive impact on team resilience by fostering shared positive emotions. Positive emotional displays are also characteristic of resilient teams (Algoe & Fredrickson, 2011). Through the top-down process of socialisation, individuals might start to comply with team norms and behaviours and converge to the team's affective tone (Collins, Lawrence, Troth, & Jordan, 2013). Therefore, resilient teams that tend to draw on positive collective emotions might foster a positive affective state and elicit individual resilience among team members (Fredrickson, 2001).

However, in certain situations, individual resilience may also inhibit team resilience and vice versa. According to the conservation of resources theory (Hobfoll, 1989), when individuals face significant demands at work, they may disengage from the work context in order to conserve resources and protect their own resilience. This disengagement may result in less effort and reduce the amount of support provided to other team members, thus lessening team identification, which may, in turn, weaken team resilience.

Ultimately, resilience at the individual and team levels may act as the micro-foundation of organisation-level resilience, as individuals and teams engage in practices that foster the development of organisational resilience (Barton & Kahn, 2018; Kahn et al., 2018). The extent to which organisations build resilience in the face of adversity may depend on the structures and routines developed at lower levels of the organisation (Barton & Kahn, 2018). For this reason, organisations may want to understand the interplay between resilience at different levels of analysis to implement strategies that ensure that individual or team resilience does not inhibit resilience at the organisational level. In other words, future research might seek to understand the impact of individual and team resilience on the development of resilience at the organisational level.

Limitations

Our systematic review is not without its limitations. As we have focused on reviewing empirical work published in ISI-listed journals for examining antecedents and outcomes of resilience, we may have excluded valuable findings published in books, research reports, and white papers (e.g., Maynard & Kennedy, 2016; Vanhove et al., 2015). Further, we only included literature in our systematic review that explicitly referred to resilience or a derived term in the text. As such, we did not include articles that dealt with related constructs, such as hardiness, which may limit the breadth of our review. However, we are confident that we identified key empirical work through our extensive screening process. Moreover, even though some of the studies we reviewed suggested causal relationships, much of the empirical quantitative evidence we reviewed was based on cross-sectional or correlational data. Therefore, we have to be cautious when making causal inferences.

CONCLUSION

Resilience is an important phenomenon, especially in today's volatile business environment. From a practical perspective, the present review has identified myriad positive workplace outcomes associated with resilience. Also, we have identified important antecedents that managers may want to foster in order to build resilience among individuals and teams. As such, our research goes beyond conceptual work on resilience in the workplace (e.g., Fisher et al., 2018) by providing a nomological framework of variables to which resilience is related at the individual and team level. Further, by highlighting opportunities for theoretical and empirical advancement of the literature, we believe the present study provides an important basis from which researchers can design studies that expand our knowledge of how resilience develops at the individual and team levels of analysis, and of the mechanisms and conditions that predict how resilience influences work outcomes.

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